Afternoon session:

Alignment Scores:-

1) Content Based Attention → Cosine similarity.
2) Additive Attention → tanh.
3) location-based → Softmax
4) General Attention → score
5) Dot product → "
6) Scaled dot product → "

Numericals:-

encoder hidden states (three encoder time steps)

$h_1 = [1, 0, 1]$

$h_2 = [0, 1, 1]$

$h_3 = [1, 1, 0]$

Decoder previous hidden state

$S_{t-1} = [1, 0, 1]$

$S_i^T h_i$

then

$$a_{i,i} = \frac{\exp(e_{ij})}{\Sigma_t}$$

Calculating vector matrix.

$$\alpha_{t,j} = \frac{\exp(\text{score}(S_{t-1}, h_j))}{\sum_{j=1} \exp(\text{score}(S_{t-1}, h_j))}$$

$\alpha_{t,1} = \exp(\text{score}($

$\alpha_1 = \frac{7.38}{12.8} = 0.5765$

$\alpha_2 = \frac{2.71}{12.8} = 0.211$

$\alpha_3 = 2.71 / 12.8 = 0.211$

1) $\begin{bmatrix}1\\0\\1\end{bmatrix} [1\ 0\ 1]$

$1+0+1$

$= 2$ $\quad 7.38$

$\exp(2) =$

2) $\begin{bmatrix}1\\0\\1\end{bmatrix} [0\ 1\ 1]$

$\exp(1)$

$= 1 = 2.71$

3) $\begin{bmatrix}1\\0\\1\end{bmatrix} [0\ 1\ 1]$

$\exp(1)$

$= 1 = 2.71$

Summation.

$C_t = 0.5765 + 0.211 + 0.211$

0.998

dot product.

$$c_t = \sum_{j=1}^{T} \alpha_{t,j} h_j$$

$\alpha_{t,j}(1) = 0.5765$

$\alpha_{t,j}(2) = 0.211$

$\alpha_{t,j}(3) = 0.211$

0.9985

$C_1 = [0.5765 \quad 0 \quad 0.5765]$ $[1 \; 0 \; 1]$

$C_2 = [0 \quad 0.24 \quad 0.211]$ $[0 \; 1 \; 1]$

$C_3 = [0.24 \quad 0.211 \quad 0]$ $[1 \; 1 \; 0]$

$C_t = 0.7875 \quad 0.422 \quad 0.7875$

30.09.2026 Lecture: 41

TRANSFORMERS:

Attention as a database retrieval

→ Relational database.
→ Query.
→ Why primary key.
→ Best, Average, Worst Case Scenarios
→ * Hash Table
  * Binary Tree
  * Graph.

Example: Google Search can't be exact.